

# Demonstration 1 (Week 4)

COS30045 Data Visualisation: what to submit and present

| Points | Due                                           | Available               | Submission                                              |
|--------|-----------------------------------------------|-------------------------|---------------------------------------------------------|
| 7      | Sunday by 23:59 (check Canvas for exact date) | 14 Sep to 18 Oct, 23:59 | File upload, <b>one attempt only</b> (no resubmissions) |

**Scope:** on-campus demonstration of exercises and content from **Weeks 1 and 2**, i.e. **Exercise 1** (Introduction to data processing with KNIME) and **Exercise 2** (Data exploration and chart types in KNIME). Exercise 0.2 (website) and Exercise 3 (data story) are separate tasks and are not part of this demo.

## 1. What to submit on Canvas

| Item                                            | Details                                                                                                                                                                                                                                           |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Annotated .knwf file, with data included</b> | Export from KNIME (right-click the workflow, then Export). The CSV must sit in the workflow's <i>data/</i> folder, and the CSV Reader must use "Relative to: Current workflow data area". Re-import the exported file once to check that it runs. |
| <b>Class exercises</b>                          | All workflows from Exercise 1 and Exercise 2, inside the file.                                                                                                                                                                                    |
| <b>Independent analysis</b>                     | Your own extra processing, transformations or charts that answer a <b>new question</b> about the TV data set. Needed for the top rating.                                                                                                          |
| <b>GenAI declaration</b>                        | Which tools you used and what you used them for.                                                                                                                                                                                                  |

## 2. What the workflows should include

**Exercise 1: "How many TV models does each brand currently have available for sale in Australia?"**

- CSV Reader reading *tv\_2026\_02\_15.csv* with a relative path
- Column Filter keeping Brand\_Reg, SoldIn and Availability Status
- String Cleaner to fix capitalisation, then String Replacer to merge duplicate brand names (Samsung / Samsung Electronics, plus at least one other pair)
- Row Filter to remove unavailable models, and another to remove models not sold in Australia (about 76, or 75, brands remain)
- GroupBy to count models per brand, then Sorter, a bar chart and a pie chart
- CSV Writer to export the cleaned brand counts

### Exercise 2

- **Q1:** histogram of screen sizes. Try bin sizes of 10, 20 and 30, and investigate the odd sizes around 150 and 175 cm (possibly misclassified TVs).
- **Q2:** scatter plot of screen size against energy use. Expression node to convert cm to inches; convert inches to a string for a bar chart; add small (<43"), medium (44-65") and large (>66") categories with an average-energy chart.
- **Q3:** screen technology (LCD, LCD (LED), OLED). GroupBy for frequency and average size per type, then a Pivot of tech x size category using mean energy, and a grouped bar chart.

## 3. What happens in class (oral Q&A / demonstration)

Be ready to explain:

- How and why you cleaned, explored and charted the data to answer each question
- Data types: nominal vs numeric (e.g. why a bar chart needs a category column, and why a histogram won't work for screen technology)
- How you imported and exported data, and why the relative path matters
- What the data set means, using the data dictionary
- What your charts show
- **How you would build a new workflow on the spot to look at another aspect of the data**

Likely questions from Exercise 2: why mean suits Labelled Energy Consumption, and what summary would suit the Star2 rating.

## 4. How it's marked (interview rubric)

| Rating                             | What it looks like                                                                                                                                                    |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Very Good</b>                   | Explains <i>why</i> each decision was made, answers follow-up questions, suggests changes to answer different questions, and includes well-justified extra workflows. |
| <b>Good</b>                        | All class workflows present, but explanations have gaps or need prompting.                                                                                            |
| <b>Satisfactory</b>                | Can say <i>what</i> a workflow does but not <i>why</i> .                                                                                                              |
| <b>Borderline / Unsatisfactory</b> | Cannot explain the work, appears to rely on GenAI, or did not attend.                                                                                                 |

There is also a separate 5-point criterion with no description, probably marked on completeness.

## 5. Things to watch out for

- **You will only be asked about work you submit.** General knowledge with no example in your file earns nothing.
- **Keep annotations short.** Reading out long node comments during the demo will be treated as GenAI use. Write brief labels and explain the rest in your own words.
- **Check that your .knwf includes the data** by importing it back into KNIME. If it doesn't, it won't run for your tutor.